

## DESIGN AND ANALYSIS OF ALGORITHM (CS-327)

## LECTURE – 2

## Cases of Algorithm

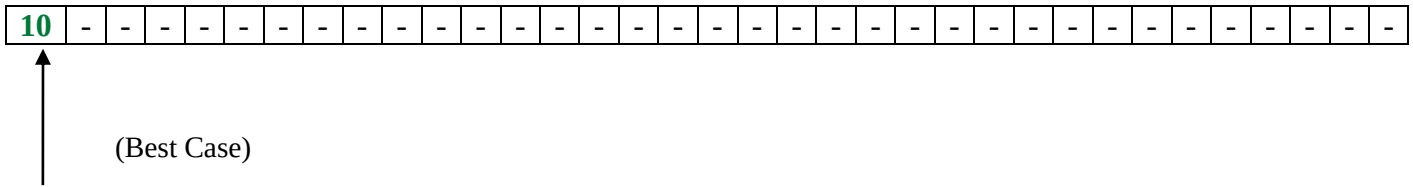
### 1. Best Case :-

While searching the key element of an array, best case is the case when we find the key element at first location of an array.

**example :-**

Size of array (N=50)

## Key Element = 10



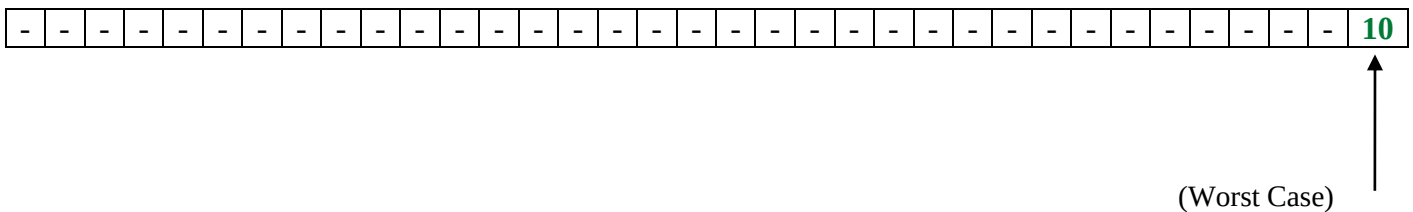
## 2. Worst Case :-

While searching the key element of an array, worst case is the case when we find the key element at last location of an array.

**example :-**

Size of array (N=50)

**Key Element = 10**



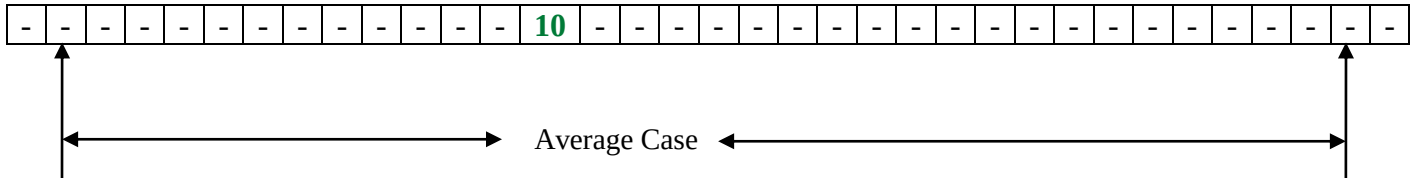
### 3. Average Case :-

While searching the key element of an array, average case is the case when we find the key element in between 2<sup>nd</sup> and 2<sup>nd</sup> last location of an array.

**example :-**

Size of array (N=50)

**Key Element = 10**



### 4. Asymptotic Case :-

Asymptotic notations was developed to provide the universal way to measure the speed and efficiency of an algorithm. Asymptotic algorithm refers for large input. we measure the behaviour of algorithm when input size reaches infinity.

**Notations:-**

$O(N)$   $\longrightarrow$  **“Big O”**  $\longrightarrow$  Upper bound  
 $O(N^2)$

$\Theta(N)$   $\longrightarrow$  **“Theta”**  $\longrightarrow$  Tight Bound  
 $n < \textcircled{3n} < 5n$

$\Omega(N)$   $\longrightarrow$  **“Omega”**  $\longrightarrow$  Lower Bound  
or  $\pi(N)$

**Note :-**

While finding the efficiency of algorithm we are always interested in worst case, which means for analyzing the complexity of algorithm we only care about the highest power and ignore the rest.

Example :-

$$\textcircled{n^5} + 4n^3 + 3n + 5$$

the ultimate complexity of above expression is  **$n^5$** .